Polar Equations and Graphs

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Recognizing polar graphs**

Identify the graph of each polar equation:

a $r = 3$ Circle of radius 3 centered at the pole.

b $\theta = \frac{\pi}{6}$ Line through the pole at angle $\frac{\pi}{6}$.

c $r = 2 + 2\cos\theta$ Cardioid (heart-shaped, symmetric about the polar axis).

d $r = 4\sin 3\theta$ Rose curve with 3 petals.

(a) Circle of radius 3 centered at the pole.; (b) Line through the pole at angle $\frac{\pi}{6}$.; (c) Cardioid (heart-shaped, symmetric about the polar axis).; (d) Rose curve with 3 petals.

2 **[Foundational]** • **Target: Recognizing polar graphs**

For the limaçon $r = 1 + 2\cos\theta$:

a Explain how you know it has an inner loop. In $r = a + b\cos\theta$, $|a| < |b|$ (here $1 < 2$) produces an inner loop.

b Find the values of θ in $[0, 2\pi)$ at which the curve passes through the pole. $1 + 2\cos\theta = 0$ gives $\cos\theta = -\frac{1}{2}$: $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$.

(a) In $r = a + b\cos\theta$, $|a| < |b|$ (here $1 < 2$) produces an inner loop.; (b) $1 + 2\cos\theta = 0$ gives $\cos\theta = -\frac{1}{2}$: $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$.

3 **[Foundational]** • **Target: Recognizing polar graphs**

For the rose $r = 5\sin 2\theta$:

a State the number of petals. $n = 2$ is even, so $2n = 4$ petals.

b State the maximum value of $|r|$ and the smallest positive θ at which it occurs. Maximum $|r| = 5$, first at $2\theta = \frac{\pi}{2}$, i.e. $\theta = \frac{\pi}{4}$.

(a) $n = 2$ is even, so $2n = 4$ petals.; (b) Maximum $|r| = 5$, first at $2\theta = \frac{\pi}{2}$, i.e. $\theta = \frac{\pi}{4}$.

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4 [Foundational] • Target: Recognizing polar graphs

Determine whether the graph of $r = 2 + \sin\theta$ passes through the pole, and state the type of limaçon.

$2 + \sin\theta \geq 1 > 0$ for all θ, so it never passes through the pole. Since $|a| > |b|$ $(2 > 1)$, it is a dimpled limaçon with no inner loop.

5 [Foundational] • Target: Symmetry tests

Test $r = 3\cos 2\theta$ for symmetry about the polar axis (replace θ with $-\theta$) and about the pole (replace r with $-r$).

Polar-axis: $\cos(2(-\theta)) = \cos(2\theta)$, unchanged — symmetric about the polar axis. Pole: replacing $r \to -r$ gives $-r = 3\cos 2\theta$, not equivalent to the original in general — the standard test is inconclusive for pole symmetry, though the curve (a 4-petal rose) is in fact symmetric about the pole by direct graphing.

6 [Foundational] • Target: Symmetry tests

Test $r = 2 + 2\sin\theta$ for symmetry about the line $\theta = \frac{\pi}{2}$ (replace θ with $\pi - \theta$).

$\sin(\pi - \theta) = \sin\theta$, so the equation is unchanged: symmetric about $\theta = \frac{\pi}{2}$ (the vertical axis).

7 [Foundational] • Target: Graphing from a table of values

For the cardioid $r = 1 + \cos\theta$, complete the table of r-values: $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$.

a $r(0) \ = 2$ **b** $r\left(\frac{\pi}{2}\right) \ = 1$ **c** $r(\pi) \ = 0$ **d** $r\left(\frac{3\pi}{2}\right) \ = 1$

(a) $= 2$; (b) $= 1$; (c) $= 0$; (d) $= 1$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Graphing from a table of values

For the circle $r = 4\sin\theta$, find the value(s) of θ in $[0, \pi)$ where r reaches its maximum, and state that maximum value.

Maximum of $\sin\theta$ is 1 at $\theta = \frac{\pi}{2}$, giving $r = 4$.

9 [Intermediate] • Target: Rose curves in depth

For the rose $r = 3\cos 5\theta$, state the number of petals and the length of each petal (maximum $|r|$).

$n = 5$ is odd, so there are $n = 5$ petals, each of length $|r|_{\max} = 3$.

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10 [Intermediate] • Target: Rose curves in depth

Explain, using the general rule for rose curves $r = a\cos(n\theta)$ or $r = a\sin(n\theta)$, why n odd gives n petals but n even gives $2n$ petals.

When n is odd, tracing θ from 0 to 2π retraces each petal a second time (due to negative r values overlapping earlier petals), so only n distinct petals appear. When n is even, no such overlap occurs, and all $2n$ petals from the full $[0, 2\pi)$ sweep are distinct.

11 [Intermediate] • Target: Identifying limaçon subtypes

Classify each limaçon $r = a + b\cos\theta$ as having an inner loop, being dimpled, or being convex (roughly oval, no dimple), based on the ratio $\frac{a}{b}$:

a $r = 1 + 3\cos\theta$ $\frac{a}{b} = \frac{1}{3} < 1$: inner loop.

b $r = 3 + 2\cos\theta$ $1 < \frac{a}{b} = \frac{3}{2} < 2$: dimpled.

c $r = 4 + \cos\theta$ $\frac{a}{b} = 4 > 2$: convex (no dimple).

(a) $\frac{a}{b} = \frac{1}{3} < 1$: inner loop.; (b) $1 < \frac{a}{b} = \frac{3}{2} < 2$: dimpled.; (c) $\frac{a}{b} = 4 > 2$: convex (no dimple).

12 [Intermediate] • Target: Finding maximum and minimum r-values

For the limaçon $r = 3 + 2\sin\theta$, find the maximum and minimum values of r, and the angles at which they occur.

Maximum $r = 5$ at $\sin\theta = 1$ ($\theta = \frac{\pi}{2}$). Minimum $r = 1$ at $\sin\theta = -1$ ($\theta = \frac{3\pi}{2}$).

13 [Intermediate] • Target: Comparing sine- and cosine-based polar curves

Explain how the graph of $r = 3 + 3\sin\theta$ compares to $r = 3 + 3\cos\theta$ (same cardioid shape, different orientation).

Both are cardioids of the same size and shape. The cosine version $r = 3 + 3\cos\theta$ is symmetric about the polar axis (horizontal), pointing right, while the sine version $r = 3 + 3\sin\theta$ is symmetric about $\theta = \frac{\pi}{2}$ (vertical), pointing up — a $90°$ rotation of one relative to the other.

14 [Intermediate] • Target: Sketching from key features

Without fully graphing, list the key features you would compute first to sketch $r = 2 - 4\cos\theta$ (max/min r, pole crossings, symmetry).

Type: limaçon with inner loop ($|a| = 2 < |b| = 4$). Symmetry: about the polar axis (cosine-based). Max $|r| = 6$ at $\theta = \pi$; min value $r = -2$ at $\theta = 0$. Pole crossings: $2 - 4\cos\theta = 0 \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}, \frac{5\pi}{3}$ (marking where the inner loop begins/ends).

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: A capstone graphing problem** • **Cross-Lesson: 2.02 Radian Measure and Arc Length**

For the rose $r = 4\cos(3\theta)$, find: (a) the number of petals, (b) the maximum value of r, (c) the value(s) of $\theta \in [0, \pi)$ where the first petal reaches its maximum length, and (d) the angles where the curve passes through the pole in $[0, \pi)$.

(a) $n = 3$ is odd: 3 petals. (b) Maximum $|r| = 4$. (c) $\cos(3\theta) = 1 \Rightarrow 3\theta = 0 \Rightarrow \theta = 0$ (first petal's tip). (d) $\cos(3\theta) = 0 \Rightarrow 3\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2} \Rightarrow \theta = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$.

16 **[Challenging]** • **Target: Distinguishing curve families by equation form alone** • **Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions**

Without graphing, state whether each polar equation is a circle, a line, a rose, or a limaçon, based purely on its algebraic form: $r = 7$, $r = 5\sin(4\theta)$, $\theta = \frac{2\pi}{5}$, $r = 2 - 2\cos\theta$.

$r = 7$: circle (constant r). $r = 5\sin(4\theta)$: rose ($n = 4$ even, 8 petals). $\theta = \frac{2\pi}{5}$: line through the pole (constant θ). $r = 2 - 2\cos\theta$: cardioid (a special limaçon, coefficient equals constant).

17 **[Challenging]** • **Target: Graphing from a table of values** • **Cross-Lesson: 2.04 Law of Sines and Law of Cosines**

Two lines have slopes $m_1 = -1$, $m_2 = -1$ and y-intercepts $b_1 = -2$, $b_2 = -2$. Classify the system.

Different slopes give exactly one intersection (independent). Equal slopes with equal intercepts give the same line (dependent, infinite solutions). Equal slopes with different intercepts give parallel lines (inconsistent, no solution). Final Answer: Consistent and dependent (infinitely many solutions)

18 **[Challenging]** • **Target: Rose curves in depth** • **Cross-Lesson: 3.01 The Unit Circle and Exact Values**

For $f(x) = 2x^3 + 2x^2 - x + 6$, which values are possible rational roots (by the Rational Root Theorem)?

Possible rational roots are $\pm\frac{p}{q}$ where p divides the constant term 6 and q divides the leading coefficient 2: $\pm\frac{1}{2}, \pm 1, \pm\frac{3}{2}, \pm 2, \pm 3, \pm 6$.

19 **[Challenging]** • **Target: Identifying limaçon subtypes** • **Cross-Lesson: 3.02 Fundamental Trigonometric Identities**

Find the constant term (independent of x) in the binomial expansion of $\left(2x^2 - \frac{1}{x}\right)^9$.

Step 1: Write the general term T_{k+1}:
$T_{k+1} = \binom{9}{k}(2x^2)^{9-k}\left(-\frac{1}{x}\right)^k = \binom{9}{k}2^{9-k}x^{2(9-k)}(-1)^k x^{-k} = \binom{9}{k}2^{9-k}(-1)^k x^{18-3k}$. Step 2: Find k for the term independent of x: Set exponent equal to 0: $18 - 3k = 0 \Rightarrow 3k = 18 \Rightarrow k = 6$. Step 3: Evaluate coefficient at $k = 6$: $\binom{9}{6} = \binom{9}{3} = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$. .

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20 [Challenging] • Target: Finding maximum and minimum r-values • Cross-Lesson: 3.03 Sum and Difference Formulas

Find $\lim_{x\to 25} \frac{\sqrt{x}-5}{x-25}$. (attempt 3)

Multiply by the conjugate: $\frac{\sqrt{x}-5}{x-25} = \frac{1}{\sqrt{x}+5} \to \frac{1}{2\cdot 5} = \frac{1}{10}$. Final Answer: $\frac{1}{10}$ Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Circles and Lines in Polar Form

What is the graph of $r = a$ where a is a constant?

a. A circle centered at the origin with radius $|a|$ ✓

b. A line through the origin

c. A parabola

d. An ellipse

$r = a \Rightarrow \sqrt{x^2+y^2} = a \Rightarrow x^2+y^2 = a^2$, a circle centered at $(0, 0)$.

22 [Foundational] • Target: Circles and Lines in Polar Form

What is the graph of $\theta = \alpha$ where α is a constant angle?

a. A straight line passing through the origin with slope $\tan\alpha$ ✓

b. A circle

c. A ray only

d. A spiral

Allowing all positive and negative values of r along the angle α produces a complete straight line through the origin.

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23 [Foundational] • Target: Circles and Lines in Polar Form

Describe the circle given by $r = -6\sin\theta$.

a. **Centered at** $(0, -3)$ **with radius** 3 ✓

b. Centered at $(0, 3)$ with radius 3

c. Centered at $(0, 0)$ with radius 6

d. Centered at $(-3, 0)$ with radius 3

Multiply by r: $r^2 = -6r\sin\theta \Rightarrow x^2 + y^2 = -6y \Rightarrow x^2 + (y+3)^2 = 9$. Center is $(0, -3)$, radius 3.

24 [Foundational] • Target: Limaçons and Cardiods

Under what condition does the limaçon $r = a + b\cos\theta$ $(a, b > 0)$ form a cardioid (heart-shaped curve)?

a. $a/b = 1$ **(i.e.** $a = b$**)** ✓

b. $a/b < 1$

c. $1 < a/b < 2$

d. $a/b \geq 2$

When $a = b$, the curve passes through the origin at a cusp, creating a cardioid.

25 [Foundational] • Target: Limaçons and Cardiods

Which equation produces a limaçon with an inner loop?

a. $r = 2 + 3\cos\theta$ ✓

b. $r = 3 + 3\cos\theta$

c. $r = 4 + 2\cos\theta$

d. $r = 3 + 2\cos\theta$

An inner loop occurs when $a < b$ (ratio $a/b < 1$). For $r = 2 + 3\cos\theta$, $a/b = 2/3 < 1$.

26 [Foundational] • Target: Limaçons and Cardiods

What type of curve is $r = 5 + 2\sin\theta$?

a. **A convex (dimple-free) limaçon** ✓

b. A cardioid

c. A limaçon with an inner loop

d. A dimpled limaçon

Here $a/b = 5/2 = 2.5 \geq 2$. When $a/b \geq 2$, the limaçon has no dimple and is convex.

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27 [Foundational] • Target: Rose Curves

How many petals does the rose curve $r = 4\cos(3\theta)$ have?

a. 3 **petals** ✓

b. 6 petals

c. 4 petals

d. 12 petals

For $r = a\cos(n\theta)$, if n is odd, the curve has exactly n petals. Here $n = 3$, so it has 3 petals.

28 [Foundational] • Target: Rose Curves

How many petals does the rose curve $r = 2\sin(4\theta)$ have?

a. 8 **petals** ✓

b. 4 petals

c. 16 petals

d. 2 petals

If n is even, the curve has $2n$ petals. Here $n = 4$ (even), so it has $2(4) = 8$ petals.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Rose Curves

What is the maximum length of each petal in $r = 5\cos(2\theta)$?

a. 5 ✓

b. 10

c. 2.5

d. 2

The maximum value of $|r| = |5\cos(2\theta)|$ is 5, which determines the tip length of each petal.

30 [Intermediate] • Target: Lemniscates and Spirals

What figure-eight-shaped curve is given by $r^2 = 9\cos(2\theta)$?

a. **A lemniscate** ✓

b. A cardioid

c. A rose curve

d. A spiral of Archimedes

Equations of the form $r^2 = a^2\cos(2\theta)$ or $r^2 = a^2\sin(2\theta)$ represent lemniscates (infinity-symbol / figure-eight curves).

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31 [Intermediate] • Target: Lemniscates and Spirals

What type of curve is represented by the polar equation $r = 2\theta$ for $\theta \geq 0$?

a. **Spiral of Archimedes** ✓

b. Logarithmic spiral

c. Lemniscate

d. Circle

Equations where radius is directly proportional to angle ($r = a\theta$) are Archimedean spirals.

32 [Intermediate] • Target: Lemniscates and Spirals

Which symmetry test verifies that a polar graph is symmetric with respect to the polar axis (the horizontal axis)?

a. **Replacing** (r, θ) **with** $(r, -\theta)$ **yields an equivalent equation** ✓

b. Replacing (r, θ) with $(-r, \theta)$

c. Replacing θ with $\theta + \pi$

d. Replacing r with $1/r$

Symmetry across the polar axis corresponds to reflection across $\theta = 0$, so replacing θ with $-\theta$ must leave the equation unchanged.

33 [Intermediate] • Target: Rose Curves

Consider the complex number $z = \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^{10}$. If z is expressed in polar form as $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $0 \leq \theta < 2\pi$, find the exact values of r and θ.

a. $r = 1, \theta = \frac{5\pi}{3}$ ✓

b. $r = 1, \theta = \frac{4\pi}{3}$

c. $r = 10, \theta = \frac{5\pi}{3}$

d. $r = \sqrt{3}, \theta = \frac{4\pi}{3}$

First, express the base complex number in polar form. Let $w = \frac{\sqrt{3}}{2} + \frac{1}{2}i$. The modulus of w is $|w| = \sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{3}{4} + \frac{1}{4}} = \sqrt{1} = 1$. The argument of w is $\arg(w) = \arctan\left(\frac{1/2}{\sqrt{3}/2}\right) = \arctan\left(\frac{1}{\sqrt{3}}\right)$. Since the real and imaginary parts are both positive, w is in the first quadrant, so $\arg(w) = \frac{\pi}{6}$. Thus.

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34 [Intermediate] • Target: Lemniscates and Spirals

What is the Cartesian equation of the polar curve $r = 5$?

a. x+y=25

b. $x^2 - y^2 = 25$

c. $x^2 + y^2 = 5$

d. $x^2 + y^2 = 25$ ✓

Applying Polar Coordinates principles, Squaring $r = 5$: $r^2 = x^2 + y^2 = 25$. Therefore, the verified correct option is $x^2 + y^2 = 25$.

35 [Intermediate] • Target: Recognizing polar graphs

What is the Cartesian equation of the polar curve $r = 6$?

a. $x^2 + y^2 = 36$ ✓

b. x+y=36

c. $x^2 + y^2 = 6$

d. $x^2 - y^2 = 36$

Applying Polar Coordinates principles, Squaring $r = 6$: $r^2 = x^2 + y^2 = 36$. Therefore, the verified correct option is $x^2 + y^2 = 36$.

36 [Intermediate] • Target: Symmetry tests

What is the Cartesian equation of the polar curve $r = 7$?

a. x+y=49

b. $x^2 - y^2 = 49$

c. $x^2 + y^2 = 7$

d. $x^2 + y^2 = 49$ ✓

Applying Polar Coordinates principles, Squaring $r = 7$: $r^2 = x^2 + y^2 = 49$. Therefore, the verified correct option is $x^2 + y^2 = 49$.

37 [Intermediate] • Target: Graphing from a table of values

What is the Cartesian equation of the polar curve $r = 8$?

a. x+y=64

b. $x^2 + y^2 = 8$

c. $x^2 - y^2 = 64$

d. $x^2 + y^2 = 64$ ✓

Applying Polar Coordinates principles, Squaring $r = 8$: $r^2 = x^2 + y^2 = 64$. Therefore, the verified correct option is $x^2 + y^2 = 64$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Rose curves in depth • Cross-Lesson: 3.01 The Unit Circle and Exact Values

What is the Cartesian equation of the polar curve $r = 9$?

a. $x^2 + y^2 = 81$ ✓

b. $x^2 - y^2 = 81$

c. x+y=81

d. $x^2 + y^2 = 9$

Applying Polar Coordinates principles, Squaring $r = 9$: $r^2 = x^2 + y^2 = 81$. Therefore, the verified correct option is $x^2 + y^2 = 81$.

39 [Challenging] • Target: Identifying limaçon subtypes • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

What is the Cartesian equation of the polar curve $r = 1$?

a. $x^2 + y^2 = 1$ ✓

b. $x^2 - y^2 = 1$

c. $x^2 + y^2 = 2$

d. x+y=1

Applying Polar Coordinates principles, Squaring $r = 1$: $r^2 = x^2 + y^2 = 1$. Therefore, the verified correct option is $x^2 + y^2 = 1$.

40 [Challenging] • Target: Finding maximum and minimum r-values • Cross-Lesson: 3.03 Sum and Difference Formulas

What is the Cartesian equation of the polar curve $r = 2$?

a. $x^2 + y^2 = 2$

b. $x^2 + y^2 = 4$ ✓

c. x+y=4

d. $x^2 - y^2 = 4$

Applying Polar Coordinates principles, Squaring $r = 2$: $r^2 = x^2 + y^2 = 4$. Therefore, the verified correct option is $x^2 + y^2 = 4$.

41 [Challenging] • Target: Comparing sine- and cosine-based polar curves • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

What is the Cartesian equation of the polar curve $r = 3$?

a. $x^2 - y^2 = 9$

b. x+y=9

c. $x^2 + y^2 = 9$ ✓

d. $x^2 + y^2 = 3$

Applying Polar Coordinates principles, Squaring $r = 3$: $r^2 = x^2 + y^2 = 9$. Therefore, the verified correct option is $x^2 + y^2 = 9$.

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42 [Challenging] • Target: Sketching from key features • Cross-Lesson: 4.01 Circles and Parabolas

What are the maximum and minimum values of $y = 2\sin(x) - 2$?

a. Max 1, Min -5

b. **Max 0, Min -4** ✓

c. Max -2, Min 2

d. Max 2, Min -2

Since $-1 \le \sin(x) \le 1$, y ranges from $-2 - 2 = -4$ to $2 - 2 = 0$.

43 [Challenging] • Target: A capstone graphing problem • Cross-Lesson: 4.02 Ellipses

What are the maximum and minimum values of $y = 3\sin(x) + 1$?

a. Max 3, Min -3

b. **Max 4, Min -2** ✓

c. Max 1, Min -1

d. Max 5, Min -3

Since $-1 \le \sin(x) \le 1$, y ranges from $-3 + 1 = -2$ to $3 + 1 = 4$.

44 [Challenging] • Target: Distinguishing curve families by equation form alone • Cross-Lesson: 4.03 Hyperbolas

What are the maximum and minimum values of $y = 5\sin(x) + 1$?

a. Max 1, Min -1

b. Max 7, Min -5

c. Max 5, Min -5

d. **Max 6, Min -4** ✓

Since $-1 \le \sin(x) \le 1$, y ranges from $-5 + 1 = -4$ to $5 + 1 = 6$.

45 [Challenging] • Target: Circles and Lines in Polar Form • Cross-Lesson: 5.01 Vectors: Components and Operations

What are the maximum and minimum values of $y = 4\sin(x) + 1$?

a. **Max 5, Min -3** ✓

b. Max 1, Min -1

c. Max 4, Min -4

d. Max 6, Min -4

Since $-1 \le \sin(x) \le 1$, y ranges from $-4 + 1 = -3$ to $4 + 1 = 5$.

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